Connect forces to velocity and position

LEARNING TARGET I can connect forces, velocity change, and final position.

A · KEY VOCABULARY

Chain	Net force → a → velocity and position.
Interval	Keep one start and end time throughout.
Reversal	v can cross zero while a keeps its sign.
Area	Add positive and negative pieces of Δx.

B · KEY EQUATIONS + UNITS

$$a = \Sigma F \div m$$

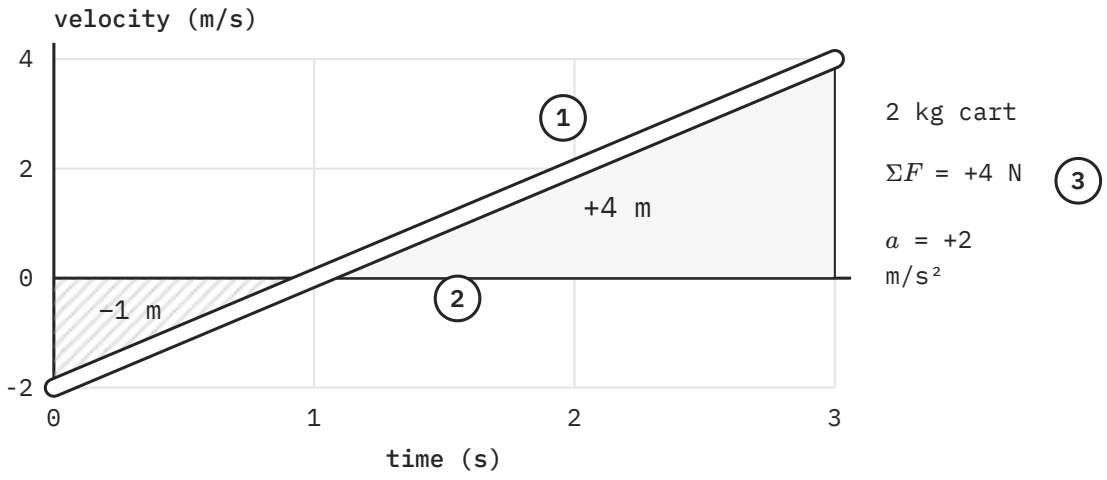
$$v_f = v_i + at$$

x_f $x_0 + v_i t + \frac{1}{2}at^2$

`model` constant a

C · COLOR IT · FORCE TO POSITION

*Velocity crosses zero at **1 s**.*



☐ **1 · PURPLE** Velocity line

☐ **2 · BLUE** Signed areas

☐ **3 · ORANGE** Net force: **+4 N**

D · WORKED EXAMPLE

A **2 kg** cart starts at **1 m** with velocity **−2 m/s**. A constant **+4 N** net force acts for **3 s**. **Find final velocity and position.**

GIVEN	EQUATION	WORK	ANSWER
m = **2 kg**; ΣF = **+4 N** x_0 = **1 m**; v_i = **−2 m/s** t = **3 s**	Find a from force. Then steady-a equations.	a = 4/2 = **+2 m/s²** v_f = −2 + 2(3) x_f = 1 − 6 + 9	**+4 m/s; +4 m** Δx = **+3 m**; distance = **5 m**.

E · THE 2026-XJ THREAD

DAY 20 OF 22 · NEW TOOL: THE FULL CHAIN

XJ's forecast needs its present motion and the forces changing it. Use one reference frame through the entire calculation.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else